

Automated Customer Review Analysis: Project Report

This report outlines the development of an end-to-end NLP pipeline designed to transform raw Amazon customer reviews into structured, actionable product recommendation articles. The system leverages state-of-the-art Transformer models for sentiment classification, unsupervised learning for product clustering, and Generative AI for content synthesis [cite: 10].

1. Project Overview & Problem Statement

With millions of reviews generated daily, manual analysis is no longer scalable for e-commerce teams [cite: 10]. This project addresses the need for automated feedback loops by processing natural language data to identify perceived quality, defects, and buyer profiles [cite: 10]. The final deliverable is a comprehensive pipeline that classifies sentiment, groups products by theme, and generates professional summaries [cite: 10].

2. System Architecture

The architecture is composed of five independent modules that can be executed on any review-based CSV dataset [cite: 10]:

- **Preprocessing:** Data cleaning, deduplication, and labeling using pandas [cite: 10].
- **Classification:** Sentiment analysis using a fine-tuned RoBERTa model (Positive, Neutral, Negative) [cite: 10].
- **Clustering:** Automated thematic grouping using TF-IDF, PCA, and K-Means [cite: 10].
- **Summarization:** Article generation via GPT-4o-mini and strategic prompt engineering [cite: 10].
- **Deployment:** User-facing web application built with Streamlit and Gradio [cite: 10].

3. Dataset Characteristics & Exploratory Data Analysis (EDA)

The analysis was performed on a dataset of Amazon product reviews, specifically focused on Electronics and Office Supplies [cite: 10].

Metric	Value
Total Cleaned Reviews	721 [cite: 10]
Unique Products	23 [cite: 10]
Average Rating	4.64 / 5.0 [cite: 10]
Sentiment Distribution	93.8% Positive · 4% Neutral · 2.2% Negative [cite: 9, 11]

Key EDA insights revealed a severe class imbalance, where the vast majority of reviews are positive [cite: 11]. This imbalance necessitates specific techniques like class-weighting during model training to ensure the system effectively identifies critical negative feedback [cite: 10, 11]. Analysis of review lengths justified a maximum sequence length of 128 for the Transformer models [cite: 6, 10].

4. Sentiment Classification: RoBERTa Fine-Tuning

To move beyond rule-based baselines like VADER, several Transformer architectures were compared [cite: 10]. RoBERTa was selected due to its superior pre-training on large-scale text data and its ability to handle nuanced language better than standard BERT [cite: 10].

Model	Type	Accuracy	F1 (Weighted)
VADER	Rule-based	~0.65 [cite: 10]	~0.61 [cite: 10]
nlptown BERT	Pretrained	~0.72 [cite: 10]	~0.70 [cite: 10]
RoBERTa Fine-tuned	Fine-tuned	~0.84 [cite: 10]	~0.82 [cite: 10]

A critical technical challenge was the "neutral" class minority (4%) [cite: 10]. This was addressed using a **Class-Weighted Loss** function, which penalizes errors in minority classes more heavily during training [cite: 10]. This technique resulted in a +6 point increase in F1-score compared to the base RoBERTa model [cite: 10].

5. Unsupervised Product Clustering

The system uses K-Means clustering to discover thematic groups without prior labels [cite: 10]. TF-IDF was used for text vectorization, followed by PCA for dimensionality reduction [cite: 10]. The optimal number of clusters (K=5) was determined using the Silhouette Score [cite: 10, 13]. The discovered categories include [cite: 7, 8]:

- Alexa Devices (e.g., Echo Plus, Echo Show) [cite: 7, 8]
- Fire Tablets (the largest cluster) [cite: 7, 8]
- E-Readers & Accessories [cite: 7, 8]
- Premium E-Readers [cite: 7, 8]
- Streaming Devices (e.g., Fire TV) [cite: 7, 8]

6. Generative AI Summarization

Using the classified and clustered reviews, GPT-4o-mini generates professional recommendation articles [cite: 10]. Each article includes an executive summary, key strengths, areas for improvement, and an ideal buyer profile [cite: 10]. This ensures that the end user receives high-level, actionable insights instead of thousands of individual text snippets [cite: 10].

7. Deployment & Accessibility

The project is deployed across three public platforms [cite: 10]:

- **Streamlit Web App:** A complete interface for real-time sentiment classification, category exploration, and article generation [cite: 10].
- **HuggingFace Hub:** Hosting the fine-tuned RoBERTa model for cloud-based inference [cite: 10].
- **Gradio Space:** An interactive demo environment for individual review testing [cite: 10].

8. Conclusions & Future Work

The pipeline successfully achieved an accuracy of ~84% [cite: 10]. Key takeaways include the power of domain-specific fine-tuning and the necessity of class-weighting in imbalanced e-commerce datasets [cite: 10]. Future iterations will focus on multilingual support and the implementation of MLOps for model drift monitoring [cite: 10].